INCH-POUND

MIL-DTL-25666/4 (USAF)
15 September 1995
SUPERSEDING
MIL-C-25973B
18 January 1971

DETAIL SPECIFICATION

CONVERTER, LIQUID OXYGEN, GCU-12A/A

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers one type of liquid oxygen converter with a capacitance gauging element.. It provides storage and pressurization of 5 liters of liquid oxygen to 70 pounds per square inch gage (psig), nominal pressure and does not include warming coils. The converter is designated type GCU-12A/A

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index for Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-V-9050	Valves, Oxygen, Pressure Relief, Aircraft
MIL-C-25516	Connectors, Electrical, Miniature, Coaxial, Environment Resistant Type, General Specification for
MIL-DTL-25666	Converter, Liquid Oxygen, Capacitance Type Gauging, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: OC-ALC/TICLA, 3001 Staff Drive, Tinker AFB, OK 73145-5990 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1660

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STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-129	Marking for Shipment and Storage
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal

(Unless otherwise indicated, copies of the above specifications, and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this documents takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Converters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Design. The converter shall conform to the dimensions and general design shown on figures 1, 2, 3, 4, and 5. The converter shall have a minimum capacity of 5 liters of liquid oxygen measured at 1 atmosphere and shall be designed for an operating pressure of 70 psig.

3.2.1 Schematic diagram. The converter shall be designed in accordance with the schematic diagram shown on figure 6.

3.2.2 Design strength. The filled converters shall be deigned to withstand the following forces without damage:

Vertical (down)	14G
Vertical (up)	6G
Lateral	6G

3.2.3 Relief valves. The relief valves shall conform to MIL-V-9050, type II.

3.2.4 Weight. The weight of the complete converter shall not exceed 12 pounds.

3.4 Marking of ports. Ports shall be marked as indicated on figure l.

3.3 Performance.

3.3.3 Final assembly pressure. The converter shall meet the performance requirements for the final assembly pressure test of MIL-DTL-25666 when subjected to a final assembly pressure of 85 psig.

3.3.4 Filling time. Filling time shall not exceed 5 minutes when the converter is tested as specified MIL-DTL-25666.

3.3.3 Evaporation loss.

3.3.3.1 Vent condition. The evaporation loss of the essentially full converter in the vent condition shall not exceed 2.5 pounds of oxygen when tested as specified in MIL-DTL-25666.

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3.3.3.2 Buildup condition. The evaporation loss of the essentially full converter in the buildup condition shall not exceed 2.5 pounds of oxygen when tested as specified in MIL-DTL-25666.

3.3.5 Buildup time. Buildup time shall not exceed 10 minutes when the converter is tested as specified in MIL-DTL-25666.

3.3.6 Capacitance. The capacitance of the sensing element of the empty converter shall be 63.5 ±0.5 picofarad (pf).

3.3.6 Inner vessel pressure.

3.3.6.1 Inner vessel. The inner vessel shall meet the performance requirements for the inner vessel pressure test specified in MIL-DTL-25666 when subjected to a pressure of 200 psig.

3.3.6.2 Permanent volumetric expansion. The converter shall meet the performance requirements for permanent volumetric expansion as specified in MIL-DTL-25666 when subjected to a pneumatic or hydrostatic pressure of 200 psig .

3 3.7 Delivery rate.

3.3.7.1 Constant flow . The converter shall deliver a flow of gaseous oxygen at a flow rate up to 72 liters per minute for not less than 45 minutes with the operating pressure (measured in the gaseous phase of the system) remaining between 70 and 80 psig after the first 15 minutes of flow. Pressures in the supply lines shall remain between 55 and 105 psig during the entire period of flow. The flow at the specified rate shall be initiated within 10 minutes after the converter has reached the operating pressure.

3.3.7.2 Cyclic flow. The converter shall deliver oxygen as specified (see 3.4.7.1) for not less than 1.5 hours when tested as specified in MIL-DTL-25666.

3.3.8 Calibration. The capacitance sensing element shall have characteristics which provide for a linear relationship between electrical capacitance and volume of liquid oxygen sensed. The relationship-shall be in accordance with figure 7, using either the equations or the referenced drawing to determine tolerances. An example of use of the equations is as follows: At the 3-liter test point, suppose the actual quantity of liquid oxygen is 3.1 liters. The measured capacitance must be between C maximum (max)) and C minimum (min).

C max = 5.9 x 3 1 + 64.1 = 82.4 pf
C min = 5.7 x 3.1 + 62.9 = 80.6 pf

3.3.8.1 Equations. The following equations shall be used to calculate the minimum and the maximum capacitances:

Values less than 1 liters	Values greater than 1 liters
C min = 5.6 L + 63.0	C min = 5.7 L + 62.9
C max = 6.0 L + 64.0	C max = 5.9 L + 64.1

C = Capacitance in PF
L = Exact quantity in liters of liquid at test point

3.3.9 Operating temperature. The operating pressure shall remain between 70 and 85 psig after the first 15 minutes of flow at the low temperature and between 65 and 80 psig after the first 15-minutes of flow at the high temperature.

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3.3.10 Vibration.

3.3.10.1 Vibration. The converter shall not exhibit physical evidence of damage after completion of the vibration tests as specified in MIL-DTL-25666.

3.3.10.2 Vibration - resonance. The converter shall not exhibit physical evidence of damage after completion of the resonance portion of the vibration test specified in MIL-DTL-25666.

3.3.10.3 Vibration - cycling. The converter shall not exhibit physical evidence of damage after completion of the cycling portion of the vibration test specified (see 4.4.1).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2)

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. The qualification inspection shall consist of all the inspection methods specified in MIL-DTL-25666.

4.3 Conformance inspection. The conformance inspection shall consist of the individual tests and the sampling plans and tests specified in MIL-DTL-25666.

4.4 Vibration.

4.4.1 Vibration - cycling. The converter shall be vibrated under the cycling conditions specified herein for the applicable periods listed in table I. The frequency shall be cycled between 5 to 50 cps and return in a 15 minute cycle at an applied double amplitude of 0.036 inch or an applied acceleration of ±4.5G which ever is the lower G value. The rate of change of frequency shall be logarithmic (see 3.3.10.3).

5. PACKAGING

5.1 Packaging. Packaging requirements shall be as specified in the contract or order (see 6.2).

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The 5 liter, 70 psig converter covered by this specification is intended for use on C-130, F-4A, C and G, and F-5A aircraft.

6..2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-25666 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from, OC-ALC/TICLA, 3001 Staff Drive, Tinker AFB, OK 73145-5990.

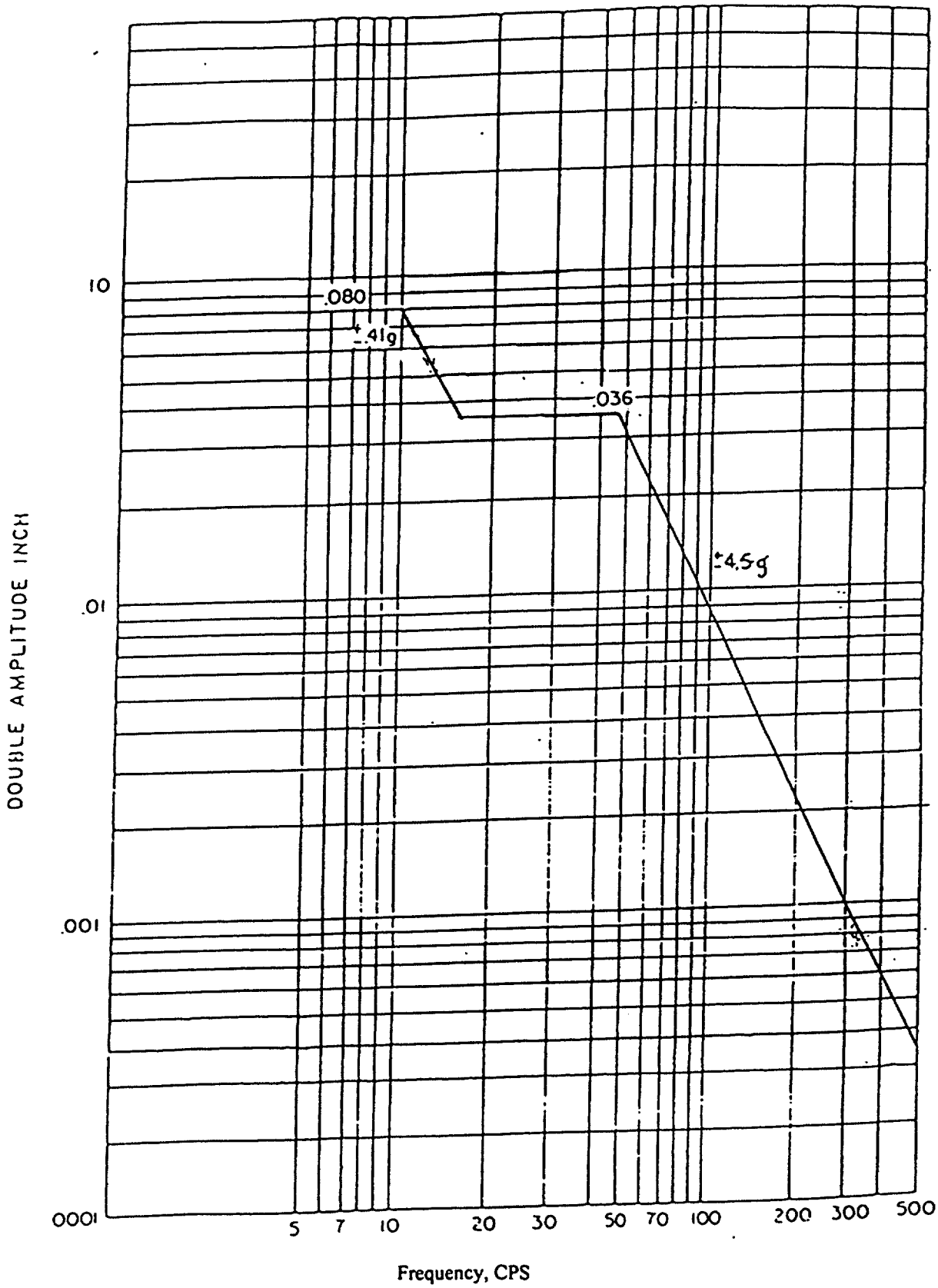
6.4 Subject term (key word) listing.

Breathing oxygen
Capacitance
Container, inner
Container, outer
Delivery rate
Dewar
Element, sensing
Evaporation loss
Gunfire
Life support
LOX
Valve, fill-buildup-vent
Valve, pressure closing
Valve, pressure relief
Vibration

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I. Range curve for converter vibration



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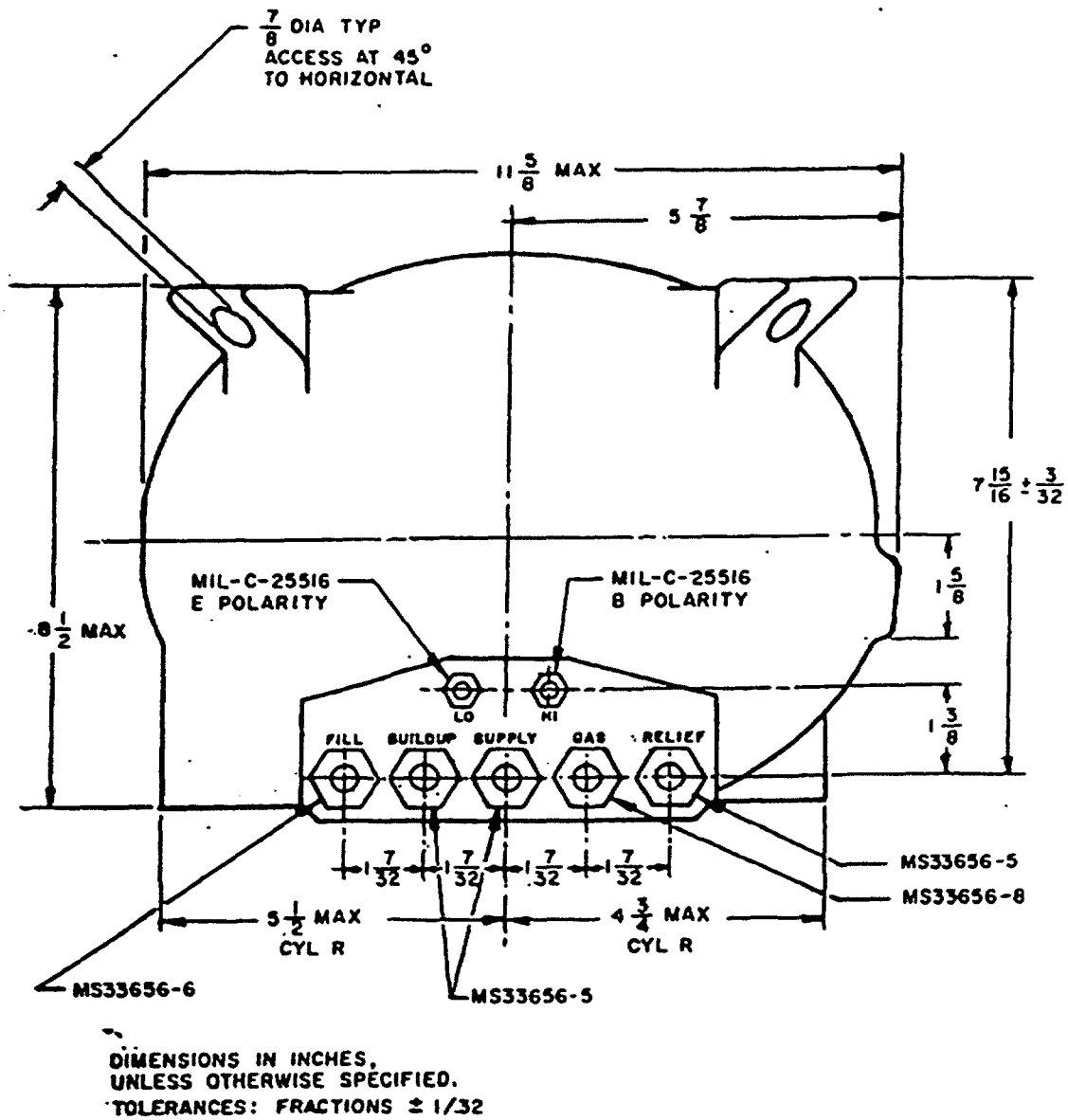


FIGURE 1. Front view

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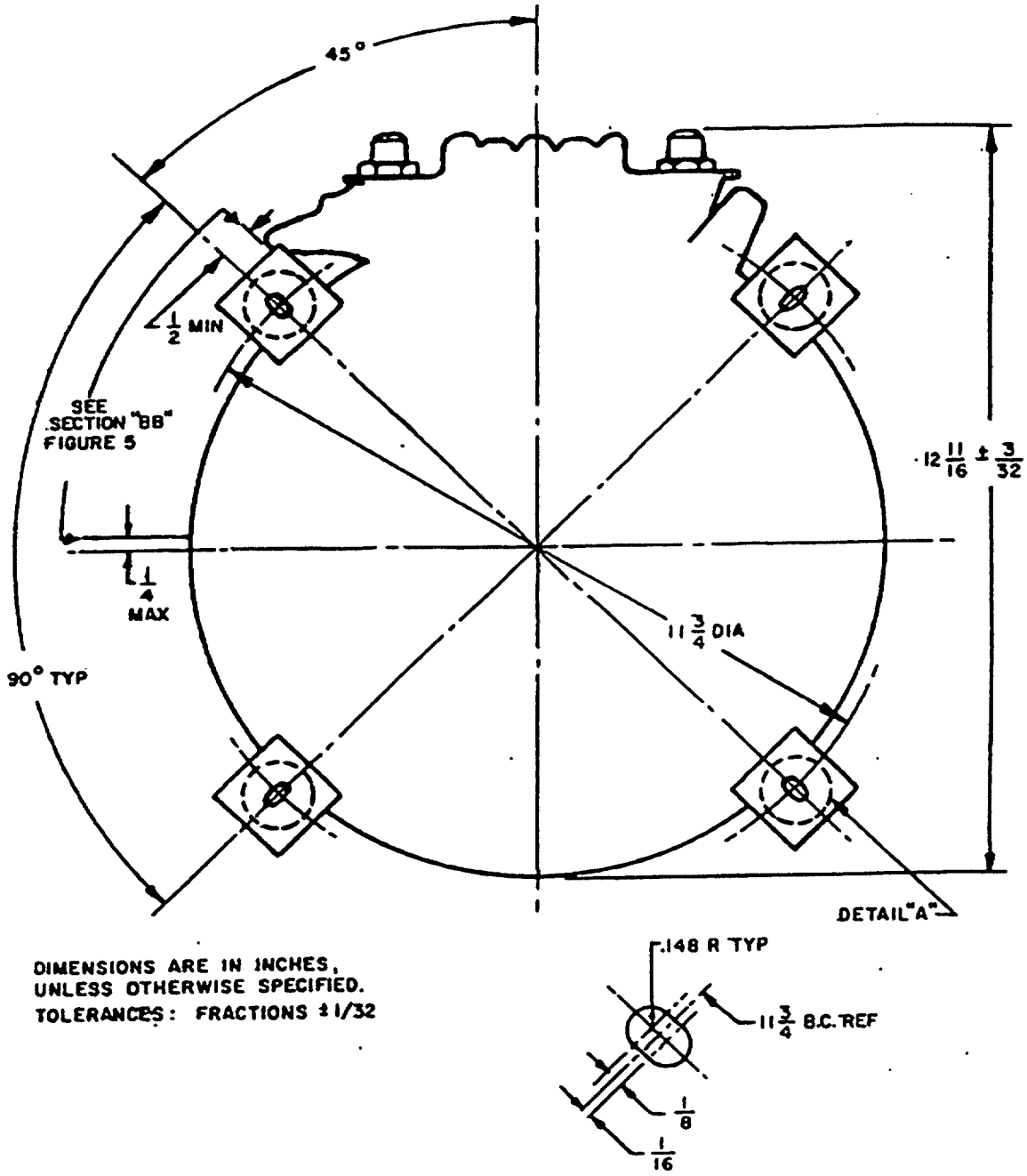


FIGURE 2. Top view

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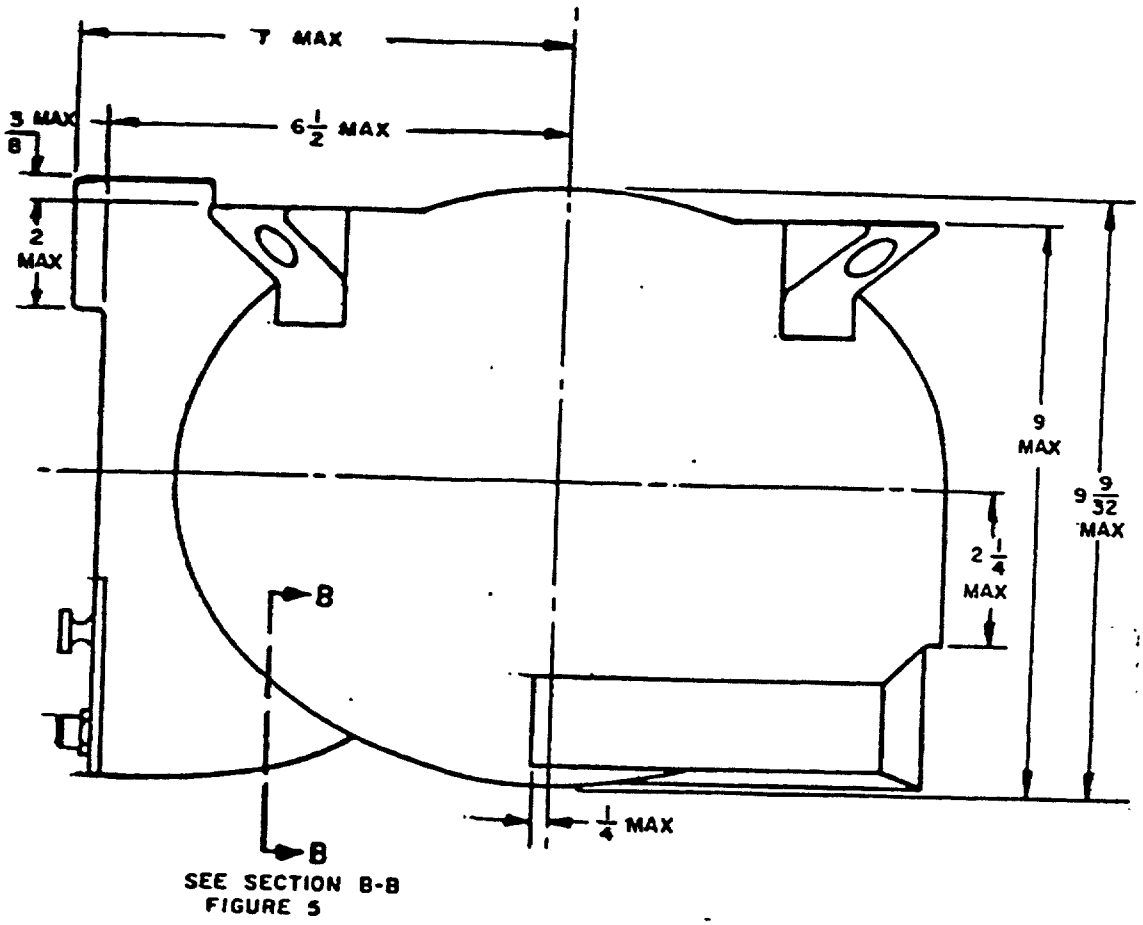
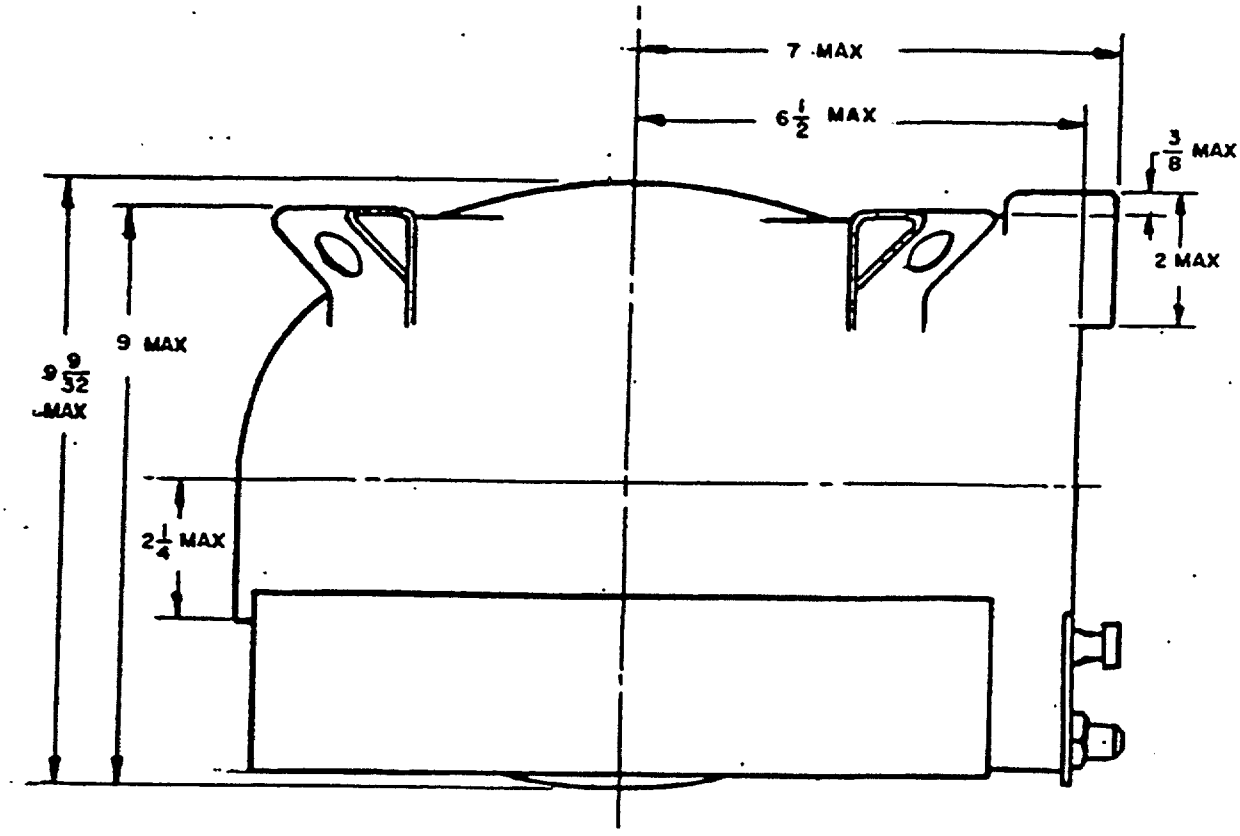


FIGURE 3. Right side

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DIMENSIONS ARE IN INCHES.
UNLESS OTHERWISE SPECIFIED.
TOLERANCE ON FRACTIONS: ± 1/32

FIGURE 4. Left side

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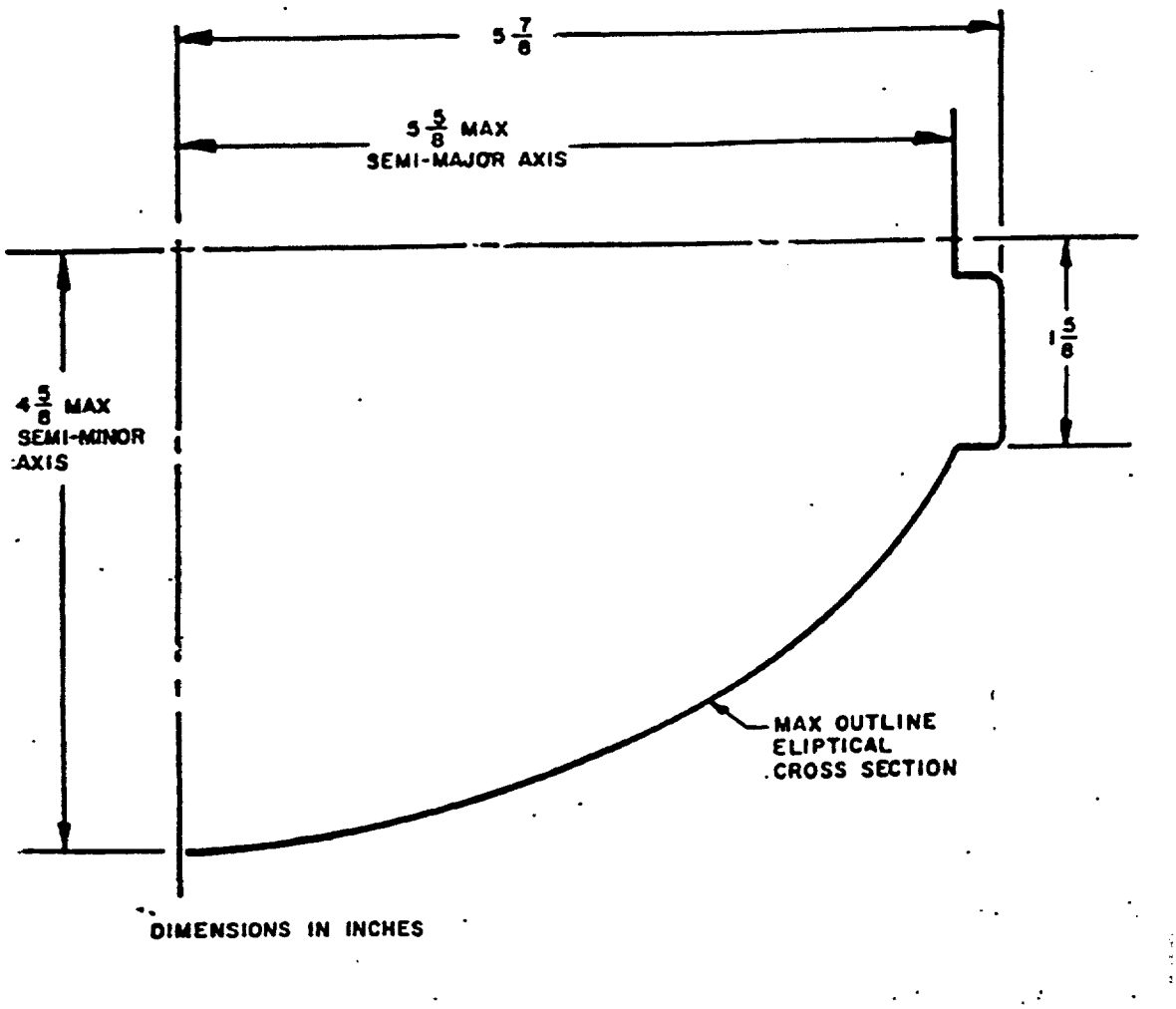


FIGURE 5. Section B-B

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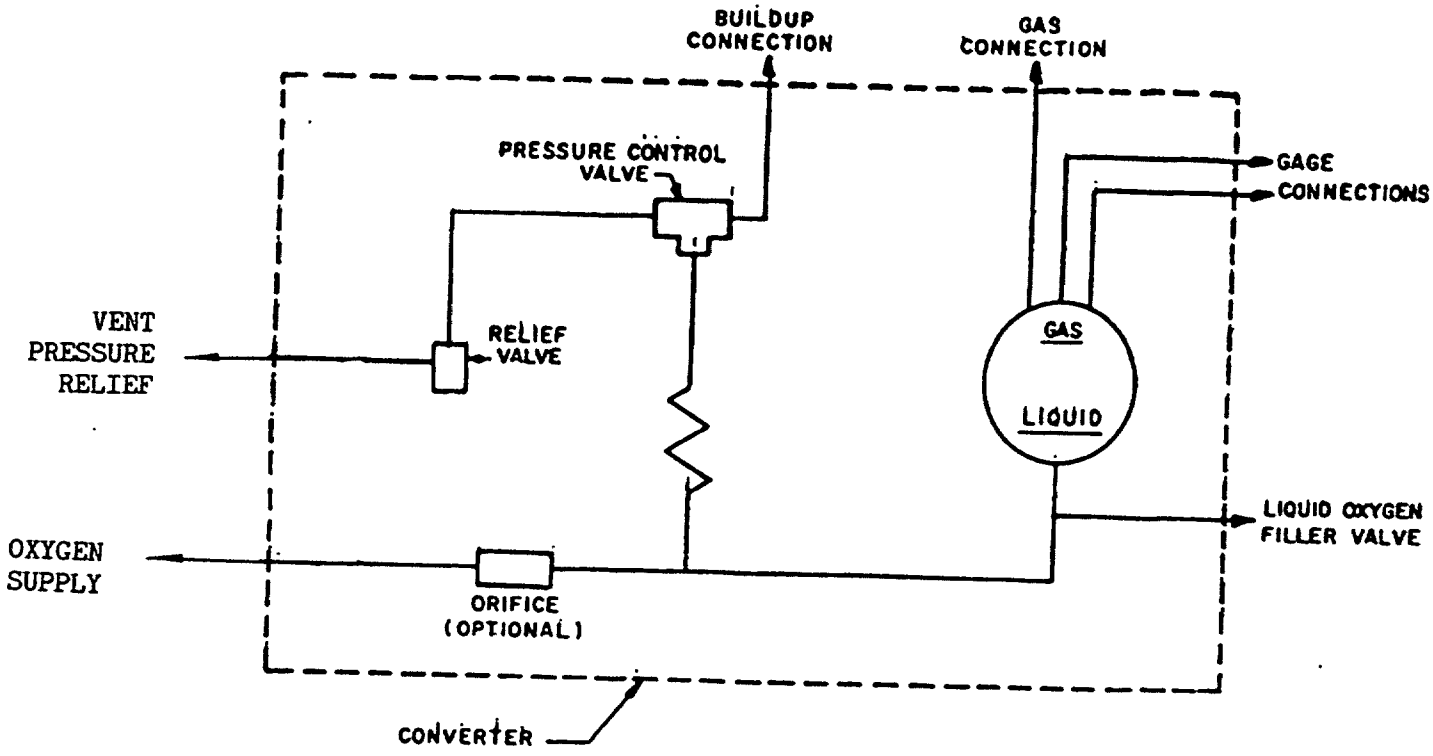
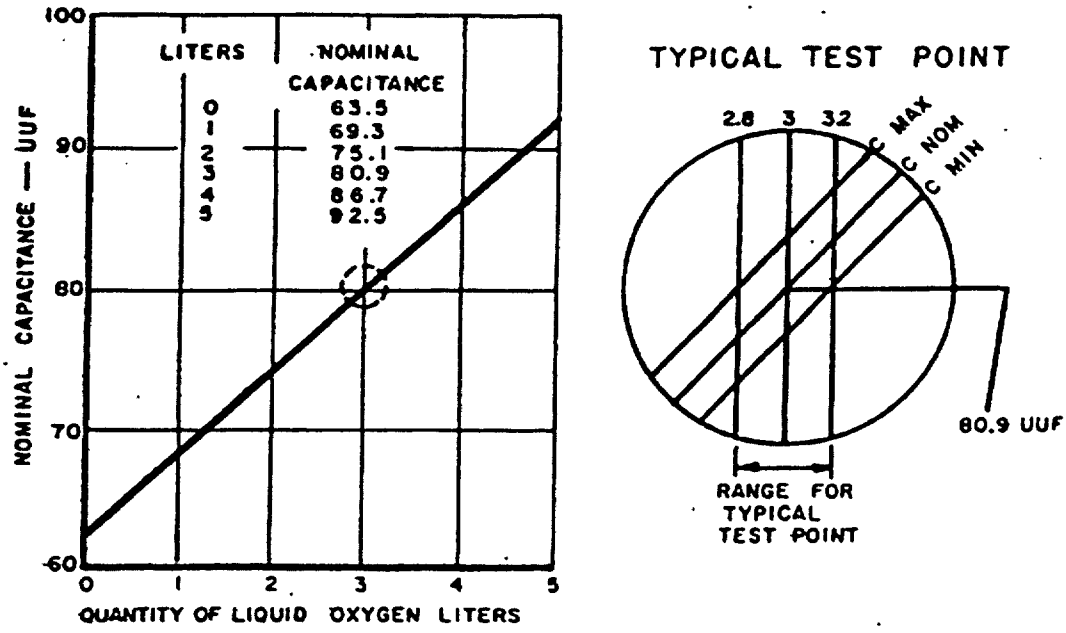


FIGURE 6. Schematic diagram

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Liquid oxygen (liters)	Nominal capacitance (pf)
0	63.5
1	69.3
2	75.1
3	80.9
4	86.7
5	92.5

FIGURE 7. Relationship between capacitance and volume of oxygen sensed

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Review activity:
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Preparing activity:
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-DTL-25666/4	950915

3. DOCUMENT TITLE: CONVERTER, LIQUID OXYGEN, GCU-12A/A

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

OC-ALC/TICLA ATTN: H. Edwards 3001 Staff Drive, Ste 1AE1/101A Tinker AFB, OK 73145-5990	b. TELEPHONE (Include Area Code) (1) Commercial: (405) 736-5960 (2) AUTOVON: 336-5960	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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